Machine Learning 1 – Exercise 3

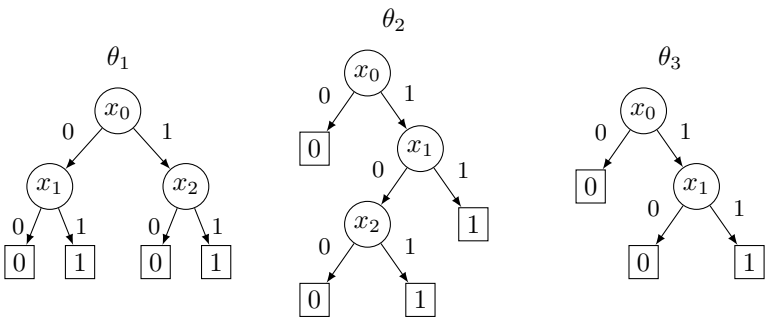
Machine Learning for Computer Vision
TU Dresden

due November 2 – 7, 2026

Deciding with binary decision trees (BDTs)

a) State the widths and depths of the three BDTs below. Which of the three BDTs defines the function $h\colon \{0,1\}^3 \to \{0,1\}$ given by the following table?

x_0	x_1	x_2	$h(x)$
0	0	0	0
1	0	0	0
0	1	0	0
1	1	0	1
0	0	1	0
1	0	1	1
0	1	1	0
1	1	1	1



b) Construct two distinct BDTs such that the function defined by these BDTs equals the function $g\colon \{0,1\}^3 \to \{0,1\}$ defined in the table below.

x_0	x_1	x_2	$g(x)$
0	0	0	0
1	0	0	1
0	1	0	0
1	1	0	1
0	0	1	1
1	0	1	0
0	1	1	1
1	1	1	0

c) Let V be a finite, non-empty set. Define an algorithm that takes any disjoint sets $A, B \subseteq \{0,1\}^V$ as the input and outputs a V-variate, $\{0,1\}$-valued BDT θ such that the function f_θ defined by this BDT satisfies $f_\theta(a) = 0$ for all $a \in A$ and $f_\theta(b) = 1$ for all $b \in B$.

d) Prove the following universality lemma: For any finite, non-empty set V and any $f\colon \{0,1\}^V \to \{0,1\}$, there exists a V-variate, $\{0,1\}$-valued BDT defining f.